

Virtual Reality Therapy Model for Treatment of Mental Disorders Associated with Gender-Based Violence

Hamida Khatri

The University of Texas at Dallas, United States

hamida.khatri@utdallas.edu

ABSTRACT

Over 35% of women experience some form of gender-based violence (GBV) in their lifetime, including intimate partner violence, rape, and more. This exposure can result in significant mental disorders within 1-5 years, manifesting as depression, post-traumatic stress disorder, and anxiety, among others, thereby affecting a staggering 58% of the world's population. The transformative power of Virtual Reality (VR) has been effectively used in treating various mental health disorders like addiction and phobias using methodologies such as cognitive behavioral therapy and exposure therapy. However, while current VR strategies are inclined towards enabling violent offenders to empathize with the sufferers and reduce aggression, there is a noticeable gap in directly catering to the therapeutic needs of GBV victims. This paper seeks to bridge the gap by introducing a novel VR Therapy Model based on human-centered design process, exclusively tailored for the rehabilitation and support for the victims of GBV.

CCS CONCEPTS

• Human-centered computing; • Applied computing; • Hardware;

KEYWORDS

virtual reality, mental disorders, gender-based violence, therapy, human-computer interaction, human-centered design

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1 INTRODUCTION

GBV is “an umbrella term for any harmful act that is perpetrated against a person’s will and that is based in socially ascribed (i.e., gender) differences between males and females” [71]. Several factors contribute to the conceptual framework of GBV, including domestic violence, physical assault, rape, sexual abuse, female genital mutilation, slavery, forced prostitution, and early or forced marriages [43, 60, 61, 64]. While men may experience GBV, in general one-third of the world’s female population experience more sexual

violence, more severe physical violence, and more control from male partners [66, 71]. By comparing the levels of vulnerability of females with males, it is known that women are eight times more frequently vulnerable to violence than men [65]. Violence is the first severe health and social consequence that GBV victims experience. Beyond the personal trauma, victims of GBV worldwide suffer a wide range of long-term psychological and mental illnesses. The prevalence of any mental disorder among victims of GBV is 24.5% as compared to men at 16.3% [57]. The spectrum of mental health disorders for victims of GBV is broad but the most frequent is depression, anxiety, post-traumatic stress disorder (PTSD), and substance abuse [12].

Women are afraid to speak about their experiences openly or share what they have been through especially to the police but when it comes to health care professionals, they have less of a hesitation when inquired [101]. According to the Maryland Coalition Against Sexual Assault (MCASA) (2013), women on average get assaulted 35 times before reporting to the police of any incident [40]. It is often the relationship between the offender and the victim that determines the likelihood of reporting the incident to the police. According to the Bureau of Justice Statistics (2013), the number of reported incidents in case that the offender is an intimate partner or former intimate partner is 25%, a friend is 18-40%, and a stranger is 46-66% [8, 9].

For GBV victims, the only accessible solution is shelter homes with at most services for short-term residence including basic necessities free of charge and daily visits by advocates for long-term shelter or housing transition [10]. It is noteworthy to say that long-term psychological support, psychiatric services, or trauma interventions do not exist at these shelter homes because of the transitory periods of stay [3]. Although there is sufficient research available on GBV, there is nominal evidence of treatments for mental disorders for women exposed to violence and limited access to trauma interventions to support them [31]. Community-based interventions, such as verbal trauma therapy, engagement and referral mediation, and recovery programs, do exist in different pockets of societies but are unapproached by victims due to negative community perception or importance of marriage privacy [44]. Studies have been conducted with randomized trials and clinical samples with several methodological limitations to assess pre- and post-treatment outcomes of mental illnesses for survivors of sexual assault including PTSD, anxiety, and depression with inconclusive results due to smaller sample size.

While existing research has begun to investigate the use of emerging technologies such as Virtual Reality (VR), Artificial Intelligence (AI), Mobile Therapy, etc. for managing and treating various mental disorders, there appears to be a gap in studies specifically targeting victims of GBV. Recognizing this, the primary aim of this research is to evaluate the effectiveness and potential of VR as a supportive

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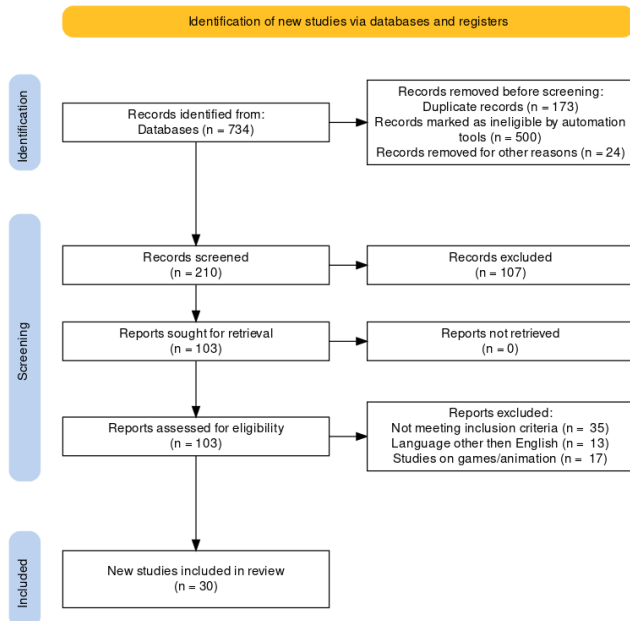


Figure 1: PRISMA flow diagram for systemic literature review [45].

tool in mitigating the mental health challenges associated with GBV. The study has two main objectives. Firstly, to comprehensively review and synthesize current literature on VR interventions aimed at mental health conditions commonly associated with GBV. Secondly, to develop a conceptual framework for a VR therapy system using a human-centered design approach tailored to the unique needs of individuals impacted by GBV. This model will serve as a foundation for future research and application in the fields of psychology and trauma-informed care.

2 METHODOLOGY

2.1 Systematic Literature Review

2.1.1 Search Strategy. For the purpose of the study, literature was identified based on mental disorders that significantly correlate with GBV victims. The search was conducted on electronic databases of PubMed, Sage Journals, ScienceDirect, International Review of Psychiatry, DigitalCommons, Frontiers of Psychology, Association for Computing Machinery, and Google Scholar starting from 1990, which is when VR research began to be used in the mental health domain, to 2023, using search terms inclusive of “virtual reality,” “virtual,” “VR,” “domestic violence,” “mental disorder,” “illness,” “cognitive exposure therapy,” “cognitive behavior therapy,” “psychosis,” “empathy,” “bullying,” “aggression,” “violence,” “gender based violence,” “intimate partner violence,” “substance abuse,” “pain,” “mental disorder,” “schizophrenia,” “trauma,” “aggression,” “intervention,” “therapy,” “anger,” and “violent behavior.”

To achieve a high standard of reporting, Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines were followed [45]. Abstracts and full texts were screened with no geographic, age, or population restrictions applied. Searches

were limited to English language only. The criteria for selection were based on (1) studies that enrolled participants in a VR related intervention, and (2) mental health issue associated with GBV as described by the Cycle of Violence by World Health Organization (WHO) (2022) [71]. 734 studies were found by cross-referencing the terms out of which 524 duplicates were removed. After screening the abstracts of 210 articles, 107 research studies were eliminated based on the exclusion criteria of no pilot studies and non-qualitative overviews. Following full-text analysis of 103 studies, it was discovered that 73 of them were not meeting the inclusion criteria. Henceforth, 30 studies were finalized to be included in the meta-analysis with the breakdown shown in Figure 1.

2.1.2 Overview of the Current Studies. GBV, as a whole, undermines safety, dignity, and health of millions of people who experience it and sabotages the mental health and well-being if not addressed in time or with proper interventions. A study conducted by Rees et al. (2011), revealed that out of a total of 2,893 women, 27.4% experienced at least one form of GBV with anxiety disorder representing 77.3%, substance abuse at 47.1%, PTSD at 56.2%, suicide attempts at 34.7%, and other mental disorders at 89.4% [11, 49]. Difficulties controlling or manipulating emotion are becoming more widely recognized as core components of a wide range of psychiatric issues. Difficulties regulating emotions can lead to unhealthy coping habits, such as drug misuse and unstable interpersonal behaviors, and can have a detrimental impact on the overall psychological well-being. For women affected by GBV, prioritizing emotional regulation may be crucial in the healing process, considering the significant emotional challenges many of them face as a consequence of such experiences. Since women are at risk of re-victimization in potential sexual or familial relationships, therapy is the obvious tool to help them distinguish between healthy and dangerous relationships [33]. Due to the lack of or minimum evidence of psychological interventions available for GBV victims, many of them find it difficult or nearly impossible to discuss their experiences with community members or service providers [44]. Reason being low perception of self-agency, fear of violence, and a sense of marital duty to keep the matter private. Regrettably, there are limited interventions that are culturally adapted and community-specific, failing to adequately address the needs of a marginalized group of victims whose experiences are not sufficiently recognized across diverse or global populations. Dialectical Behavioral Therapy (DBT) was developed to assist clients facing multiple issues, particularly those struggling with severe and chronic emotional dysregulation, and it is also aimed at treating female victims of trauma and abuse [33, 46]. While there are limited clinical reports focusing directly on enhancing emotional well-being, recent studies have demonstrated the effectiveness of cognitive-behavioral interventions in treating symptoms of PTSD. However, these studies also suggest that other dimensions of psychiatric trauma and emotional distress are often overlooked. [34].

In 2014, a GBV treatment protocol based on WHO global standards was released, and Afghanistan was the first country to develop the guidelines and adapt it countrywide [69]. The protocol encompasses various sections including survivor-centered treatment approaches, handling rape cases, treating wounds and burns, and the collection of medico-legal documentation. Additionally,

it outlines guidelines for health care providers on delivering high-quality care specifically tailored to the needs of victims of GBV. The key factors identified in this protocol are attitude (showing respect, do not judge, guarantee confidentiality, be empathetic, show patience, adjust attitude according to the characteristics of the survivor), active listening (attend to body language, listen attentively and actively, ask questions, accept physical and emotional reactions), empowerment, and resilience [69, 102]. Survivor-centered approach is a term used to describe a strategy in which the emphasis is on the victim needing help [62, 63]. This approach to treatment prioritizes the victim, placing their needs, well-being, and preferences at the center of all considerations and interventions. While there are different concepts of survivor-centered care, a respectful approach, constructive listening, and support for personal autonomy are essential to ensuring that the care provided is tailored to each victim's individual needs.

With technological progress, sectors such as healthcare, legal, and law enforcement worldwide are innovating to address sexual violence. Efforts include developing methods for forensic evidence collection, creating animated characters to raise awareness of violence and human rights among young adults experiencing GBV, utilizing gamification to offer immersive experiences from a third-person perspective to foster empathy for victims, and, most recently, deploying AI chatbots to offer accessible and immediate informational support [17, 18, 32, 37, 41]. In 2016, GBV in Nepal was reported to be 48% as noted by the government officials [72]. A group of specialists collaborated with United Nations experts to develop an innovative initiative aimed at challenging traditional gender roles through the use of gamification [7, 55, 72]. A variety of animated videos were created that focused on single acts of abuse such as harassment, domestic violence, and sexual assault. The most interesting fact was that gender roles were reversed so that the abuser is made aware of the experiences that the abused goes through when they are violated physically, emotionally, sexually, or verbally. This research method encouraged healthy conversations and raised questions about gender norms [72]. In 2019, the prevalence of GBV was extremely high in Thailand such that it was reported by the Thailand Development Research Institute (TDRI) to have 30,000 rapes per year and 87% unreported cases overall [62]. To counter this issue, an AI-powered chat application called "Sis Bot" was introduced to provide legal support to the victims [62, 63]. Enliven Media, in partnership with the Dutch Ministry of Justice, is addressing domestic violence through a VR project named "Don't Forget Me" [28, 30]. This project is designed to make offenders confront the consequences of their actions on the victims, highlighting the emotional and psychological impact. By utilizing VR technology, the initiative seeks to bridge the emotional divide between the abuser and the abused, encouraging parents to reflect on their behavior and recognize the effects their conflicts have on their children [51, 53].

A case study written by Mary Anne Franks (2017), states that VR is a reflection of the real world used as a tool to replicate the horrors of trauma and abuse [21]. The primary goal of VR is the achievement of presence, a sense of being by magnifying effects of a user's response to a virtual stimulus that resembles a parallel response to the real world. The emergence of VR as a tool for investigation, assessment, and management of medical and psychiatric disorders

have been witnessed in the last few years [54]. From social anxiety to debilitating pain and Alzheimer's disease, a recent generation of psychiatric science is pioneering VR to identify and cure medical disorders. While many of these innovative approaches are still in the research phase within laboratory settings, some are beginning to be integrated into the practices of therapists, making their way into clinical and therapeutic environments. VR content is particularly designed to support exposure therapy, a treatment approach where individuals are exposed to stimuli in a controlled and safe environment. This method allows them to gradually learn that the threats they fear are not as dangerous as they perceive, helping to alleviate their concerns and anxiety [15]. For instance, in the case of a person with PTSD, VR tools enable therapists to facilitate imagery exposure therapy sessions. Through VR, therapists can recreate a safe and controlled environment where the patient can revisit their traumatic memories. This approach allows individuals to confront and process their trauma within a secure setting, aiding in their recovery process.

Oxford VR Professor Daniel Freeman in 2018 published a review to assess the use of VR and understand how it can treat mental health conditions [22]. In over 25 years with over 285 trials, the findings explicitly confirm that VR is a verified medium in delivering rapid and lasting improvements to mental health patients. In 2018, Oxford VR conducted the largest randomized controlled trial involving 100 participants who had suffered from a fear of heights for an average of 30 years, by providing them with VR therapy. This innovative approach utilized VR to expose participants to scenarios that safely confronted their phobias, aiming to reduce their fear in a controlled and therapeutic manner. The research consisted of 30-minutes of interventions in a clinical setting with an assisted guide for each of the five treatment sessions. The results of the trial were highly significant than the use of regular face-to-face therapy. A decline in their fear of heights was seen by all participants in the study, with an average reduction of 68% [22]. Half of the participants saw a decline of over 75% in their anxiety levels [22].

One of the breakthrough implementations of VR in the mental health space is of "gameChange," a National Institute for Health and Care Research (NIHR) i4i Mental Health Challenge Award study delivered by the U.K. National Health Service (NHS) that offers a state-of-the-art VR simulation to patients with psychosis and anxiety issues [94, 100]. The sessions entailed a new automated VR cognitive treatment (VRCT) plan that was developed by using a socially inclusive design process. The research aimed to identify whether VR cognitive treatment led to reduced avoidance and distress to real world situations, and psychiatric symptoms of paranoia, depression, and suicidal ideation [29, 70]. The treatment effects were maintained for six months and shared positive results with significant decrease in psychosis and anxiety symptoms.

As part of the larger program of work done by the experts and researchers at gameChange, another study was conducted to use VRCT for patients with psychosis that used avatar coach such that the need for a therapist or VR guide is not required, offering the potential of highly scalable treatment services. The study consisted of a single-blind parallel-group randomized controlled trial called THRIVE (Therapeutic Realistic Immersive Virtual Environments) [23]. The research design tested whether the automated VRCT will reduce the persecutory delusions of patients more effectively than

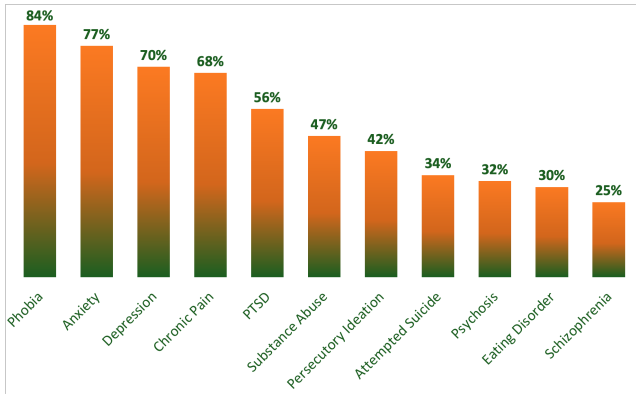


Figure 2: Mental Disorders Associated with GBV [15, 22–25, 63, 75].

VR mental relaxation (VRMT) techniques [14]. VRCT aimed to test participants’ fear expectations around other people in order to relearn for safety measures and thus minimize the delusion. The VRMT was designed with the goal of helping participants achieve a calmer state through the use of relaxation techniques. The aim was to lessen their anxiety regarding social interactions with others, making them feel more at ease in various social settings.

For the last 25 years, VR has been applied with promising results to understand and treat mental health disorders. It has become a tool for therapists to use. THRIVE trail is the first study conducted to automate the process of therapy for mental healthcare. The expansive potential of immersive architecture within VR for narrative storytelling offers significant opportunities in treating a wide range of mental health issues encountered by victims of GBV. This innovative approach can provide a therapeutic environment for those who may not even be aware of their psychological conditions, facilitating a unique and impactful path to healing through engaging and personalized narratives. VR-based therapy has the potential to assist in recognizing various disorders and can be instrumental in treating them over time. This technology offers a controlled environment where individuals can safely confront and work through their issues, making it a valuable tool in the therapeutic process.

2.1.3 Snapshot of Mental Health Disorders Associated with GBV. By meticulously examining the existing literature on the potential of VR interventions for various mental health disorders that are similarly faced by GBV victims, the profound scrutiny of data reveals that these individuals often face a complex array of mental health challenges as illustrated in Figure 2. Notably, on the day of assault, 47% of these victims resort to drug use, and there is a pronounced prevalence of alcohol use disorders. The possibility of anxiety and the threat of suicide loom large, with abused women being over three times more susceptible. Depression, a pervasive ailment, is diagnosed in a staggering 35%-70% of these victims [70]. Intriguingly, schizophrenia not only exacerbates the risk of victimization but also presents a 25% lifetime prevalence of rape [99]. A significant 30% of individuals diagnosed with eating disorders bear the scars of sexual abuse, and the prevalence of traumatic events in this demographic alternate between 37%-100% [13]. The pervasive nature of sexual

Disorder	Therapy	Result(s)
Phobia (fear)	Cognitive Behavioral Therapy	Significant ↓ in levels of fear. Significant ↑ in self-compassion.
Anxiety	Mindfulness-Based Stress Reduction Therapy & Exposure Therapy	Significant ↓ in anxiety symptoms.
Depression	Cognitive Behavioral Therapy	Significant ↓ in depression severity and self-criticism. Significant ↑ in self-compassion.
Chronic Pain	Exposure Therapy	Significant ↓ in levels of subjective pain.
PTSD	Exposure Therapy	Significant ↓ across a range of symptoms (anger, depression, etc.) with improved social interactions.
Substance Abuse	Exposure Therapy	Significant ↓ in craving for alcohol and drug use.
Persecutory Ideation	VR Cognitive Treatment & VR Mental Relaxation Treatment	Significant ↓ in delusional conviction.
Attempted Suicide	Prolonged Exposure & Virtual Reality Exposure Therapy	Significant ↓ in suicidal ideation.
Psychosis	VR Cognitive Treatment	Significant ↓ in levels of distress and adverse experiences.
Eating Disorder	Cognitive Behavioral Therapy	Significant ↓ in eating disorder and body image psychopathology.
Schizophrenia	Social Skills Training & Cognitive Empathy	Significant ↓ in negative symptomatology, psychopathology, and social avoidance as well as improvement in social skill mastery.

Figure 3: Studies Assessing the Use of VR for the Treatment of Mental Disorders.

violence directly escalates fear levels in women. A compelling 84% correlation exists between persecutory ideation and violence, predominantly rooted in anger. Physical pain, often a silent testament to their ordeal, is prevalent, with 42% of victims bearing injuries resultant from partner-inflicted violence. The grim reality is further underscored by the heightened risk of attempted suicide among women who have endured sexual abuse and intimate partner violence [19, 48]. Moreover, the relationship between psychosis and violence is particularly pronounced for women, elevating their risk significantly.

2.1.4 Studies Assessing the Use of VR for Treatment of Mental Disorders. Figure 3 demonstrates the application of VR in studying populations with diverse mental disorders, showcasing significant positive outcomes. The data presented in this figure highlight how VR interventions have effectively contributed to reducing the severity of illnesses among the user/client base, indicating VR’s potential as a powerful tool in mental health treatment and rehabilitation. Substance abuse is intricately linked with traumatic experiences, with victims often resorting to substances as a coping mechanism for trauma-induced depression and other symptoms [35, 58]. This can escalate into substance use disorder (SUD), which is frequently comorbid with PTSD. Alarming, up to 59% of females with PTSD develop substance abuse issues [1, 39, 47]. Women subjected to GBV face heightened psychiatric illnesses, physical disabilities, and a diminished quality of life [50]. Shockingly, 42% of GBV victims resort to drugs post-assault, and they are 2.3 times more likely to develop alcohol disorders [42, 68]. VR’s cue reactivity exposure therapy has shown potential in reducing cravings for drugs and alcohol [73, 74]. Anxiety disorders, characterized by excessive worry, can severely hamper daily functionality for GBV victims [2]. Many avoid therapy due to stigma as mentioned earlier [44]. GBV victims, especially women, face a heightened risk of anxiety and suicide, being over three times more likely to experience these issues [68]. However, VR has shown potential in treating anxiety, especially through mindfulness techniques and CBT [75, 76]. Depression is

another significant concern, with societal stigma and feelings of powerlessness being common triggers for GBV victims [70]. A staggering 35-70% of GBV victims are diagnosed with depression [79]. VR interventions, such as the SPARX project, have shown potential in reducing depressive symptoms [77, 78]. Eating disorders are prevalent among GBV victims, with 30% having a history of sexual abuse [80]. These disorders often arise from body image disturbances and attempts to regain control [16]. VR interventions have shown promise in addressing these disorders [81, 82]. Phobias, or intense fears, can be triggered by various traumatic experiences in GBV victims [83]. VR has been effective in treating specific phobias, such as acrophobia and social phobia demonstrating its efficacy [84]. PTSD is prevalent among GBV victims, with 20% experiencing it, especially after intimate partner violence [5, 19, 26, 85]. VR interventions have shown potential in reducing PTSD symptoms [86, 87]. Schizophrenia, a severe psychiatric disorder, makes women especially vulnerable to GBV [52, 88]. VR studies have shown promise in addressing the social and empathic challenges associated with schizophrenia [89, 90, 99]. Delusion Disorder (DD), especially persecutory delusion, is prevalent among GBV victims, with an 84% association between persecutory delusion and violent behaviors [91]. VR interventions have shown potential in reducing delusional convictions. Pain, both physical and mental, is a significant concern for GBV victims, leading to various health risks [23, 24]. VR has shown promise in pain management [25, 92, 93]. Psychosis increases the risk of violence, especially for women [56, 67]. VR interventions, like those by Lambe et al. (2020) and Valmaggia et al. (2007), have shown potential in addressing the challenges associated with psychosis [94, 95]. Lastly, suicide, while more prevalent among men globally, sees a significant number of attempts by women, especially in low- and middle-income countries [69]. GBV victims, especially women, are more likely to consider suicide [97]. VR interventions, like the one by Noor et al. (2018), have shown promise in reducing suicidal ideation [96].

2.1.5 Virtual Reality Therapy Model. Since 1970, GBV has been linked to a significant increase in morbidity and mortality, especially among women. Battered women suffer greater physical and emotional damage than men, according to some studies [1, 39]. Given the insights gained from various studies and interventions, it is not surprising that a significant portion of strategies designed to prevent and treat GBV concentrate on offering treatment to offenders. This approach aims to address the root causes of abusive behavior, thereby not only assisting in the rehabilitation of the offender but also contributing to the broader goal of reducing the incidence of GBV [4, 20, 27]. To shift the focus towards supporting female victims of GBV, this segment introduces a framework for a VR-based therapeutic model. This model aims to mitigate emotional and psychological traumas while fostering self-empowerment among these women. It is enhanced by a human-centered approach that prioritizes the well-being of women who have endured such traumatic experiences. This model seeks to offer a comprehensive support system that not only addresses the immediate psychological impacts of GBV but also empowers victims to reclaim their sense of agency and autonomy. A human-centered approach prioritizes the individual as the primary focus of study, facilitating the segmentation of the population into smaller, more manageable subgroups. These subgroups

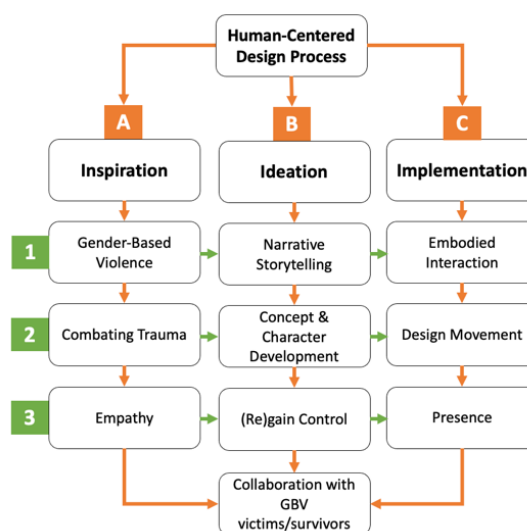


Figure 4: Virtual Reality Therapy Model Based on Human-Centered Design Process for GBV Victims.

consist of individuals who share similar experiences, allowing for a more tailored and effective response to their specific needs and circumstances. This method emphasizes understanding and addressing the unique challenges and perspectives of each person, thereby enhancing the efficacy of interventions and support provided [6]. Human-centered studies enhance the applicability and relevance of research findings across different contexts compared to variable-centered studies. This is because variable-centered approaches may not account for the uneven distribution of variations among all individuals surveyed, leading to potential discrepancies and contradictory findings in previous research. By focusing on the experiences and perspectives of individuals, human-centered studies aim to provide a more nuanced and accurate understanding of the phenomena being studied, thereby improving the reliability and generalizability of research outcomes [38].

The proposed human-centered design process for creating VR therapy simulations for victims of GBV emphasizes an iterative approach where their needs and experiences are central to every stage of the design process. This method ensures that the developed treatment is not only tailored to the victim's specific circumstances but also practical in its application. By actively involving victims in the development process, the aim is to craft VR simulations that are both empathetic to their experiences and effective in addressing the psychological impacts of GBV, ultimately offering a more personalized and impactful therapeutic experience [98]. The design process is structured to reflect the interests and needs of these victims, integrating therapeutic principles from Person-Centered Therapy (PCT) as developed by Carl Rogers. This approach is guided by Rogers' research, which emphasizes the importance of three core factors: empathy, unconditional positive regard, and congruence. By incorporating these principles, the design process seeks to create a therapeutic environment that is deeply empathetic, offers unconditional positive regard to the victims, and ensures congruence

between the therapists' external behaviors and internal feelings. This alignment with PCT principles is intended to foster a supportive and healing space that respects and validates the experiences of GBV victims, facilitating their journey towards recovery. The initial design brief, workshops, and individual sessions will address key questions by outlining the treatment, production of workflows, prototypes of the scenarios to obtain feedback, building 3D models of the virtual environments, designing and animating virtual characters, software implementation of the workflows, and extensive user testing in all stages of the design process. The three-tiered structure of human-centered architecture for the design process encompasses [A] Inspiration, [B] Ideation, and [C] Implementation as depicted in Figure 4.

The initial research phase prioritizes immersing into and understanding the lived experiences of the victims of GBV. This deep engagement is crucial for grasping the extent of trauma endured by victims and for tailoring VR as an effective therapeutic tool, leveraging empathetic insights to bridge the gap between technology and human experience. GBV is a multifaceted societal problem with both immediate and long-lasting psychological impacts, such as depression, PTSD, and substance abuse. During the ideation phase, the focus shifts to comprehending these psychological effects from the victims' viewpoints. This is achieved through storytelling and concept development, which serve as foundational elements in designing VR interventions that accurately mirror the realities of those affected by GBV. This process ensures that the VR design is informed by and reflective of the victims' actual experiences and the complexities of their trauma. In the implementation stage, the developed VR solution undergoes refinement based on continuous feedback from users, aiming to fine-tune its effectiveness in supporting GBV victims. This feedback loop is essential for ensuring that the VR intervention meets the therapeutic needs of the victims, providing a safe and controlled environment for them to confront and process their experiences. Empathy plays a critical role throughout all stages of this process, from initial research to final implementation. It serves as a guiding principle for offering support to GBV victims and shapes the interactions with professionals across sectors involved in the project. This empathetic approach ensures that the VR solutions developed are not only technologically advanced but also deeply human-centric, offering meaningful support to the victims on their journey towards healing.

The goal is to cultivate a survivor-centered methodology, integrating insights gained from interactions with victims of GBV to guide the creation of VR interventions. Storytelling emerges as a vital component in this approach, serving as a powerful tool for the victims to comprehend and articulate their experiences. This narrative element is pivotal for empowering victims, providing them with a platform to voice their stories, which in turn informs the development of VR content. Special emphasis is placed on capturing the victims' viewpoint within the VR environment, ensuring that the content not only reflects their experiences but also promotes a sense of agency and control for the user. The deployment of VR solutions is characterized by a cyclical process of testing and refinement, utilizing feedback from the victims to optimize the interventions' effectiveness and ensure they make a positive difference in the lives of the victims. Key to creating a compelling VR experience is the incorporation of embodied interaction, movement design, and

the sense of presence. These elements are crucial for crafting an immersive environment that not only engages the victim on a multisensory level but also supports their healing journey by fostering a sense of empowerment. This survivor-centered approach, with its foundation in empathetic engagement and narrative empowerment, seeks to leverage the unique capabilities of VR technology to offer a transformative tool for the recovery and empowerment of GBV victims. Through carefully designed interactions and environments, this model aims to provide a therapeutic space that respects and amplifies the voices of the victims, contributing to their process of healing, and offering a new way for them to regain control over their lives.

3 FUTURE OUTLOOK

The exploration of VR as a therapeutic intervention for victims of GBV is currently in its nascent stages, revealing only a small portion of its potential. This emerging field of study hints at a vast, uncharted territory of possibilities for using VR technology to support the healing and empowerment of GBV victims. The initial outcomes from research into the use of VR for addressing mental health issues related to GBV are encouraging, pointing towards VR's potential as a powerful therapeutic tool. Despite these promising results, there remains a considerable path forward in refining the VR therapy model and tackling the ethical and intricate challenges associated with creating simulations that touch on highly sensitive subjects. Key concerns include the risk of "traumatic coupling" where the simulation might unintentionally retraumatize victims by too closely replicating their traumatic experiences, and the broader ethics of reproducing such events within a virtual environment. These issues demand a thoughtful and nuanced approach to the development of VR content, ensuring that interventions are not only effective but also ethically responsible and sensitive to the potential impacts on the victims. As this project progresses, the imperative to develop ethically sound, effective, and compassionate VR interventions will guide the ongoing research and development efforts [36, 59]. This will involve ongoing dialogue among technologists, clinicians, ethicists, and, crucially, victims themselves, to ensure that VR therapies are developed in a manner that honors and supports the healing process of those affected by GBV.

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